Pre-FOMC Announcement Drift After 2020

Draft

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Abstract

Abstract to be written after the holdout result (task T10/T11).

1 Introduction

2 Related Literature

3 Data

4 Empirical Design

4.1 Identification

The object of interest is the return earned in the window that ends before the FOMC statement is released, in excess of the risk-free rate. Close-to-close daily returns include the announcement reaction and therefore measure a different quantity, so the primary test uses intraday prices. The comparison of interest is between meetings held in calm and in anxious states, where the state is the VIX close five trading days before the meeting (lagged one week) and the anxious threshold is fixed at the development-sample median of 15.07 VIX points. The threshold is not re-estimated on the holdout. The secondary specification replaces the indicator with the continuous lagged VIX.

Identification rests on the timing of the window and on the state being predetermined. Lagging the state by a week limits contamination from anticipation of the meeting itself. The claim is a conditional association between pre-announcement returns and the prior volatility state, not a causal effect of the Committee's actions, and the paper is framed accordingly.

4.2 Threats to inference

Episode clustering. Anxious meetings concentrate in a small number of volatility episodes, so the effective number of independent observations is far smaller than the number of meetings. We cluster by calendar year as an ex-ante proxy for episodes, use CR1 cluster-robust standard errors, and treat the wild cluster bootstrap-t as primary because the holdout has few clusters. Asymptotic t statistics are descriptive only.

Power. With a small number of independent episodes the design can detect a large premium but cannot distinguish a half-sized premium from zero. We therefore report minimum detectable effects from a clustered power simulation using realized volatility, and present non-rejections as bounds on the premium rather than as evidence of absence.

Release-time measurement. Statement release times are assigned by rule rather than observed per meeting, so the window may include a few minutes of the announcement reaction in some early meetings. This is flagged as unverified and affects the level of the estimate more than the state comparison.

Non-FOMC confounds. A premium measured on FOMC days could reflect generic high-volatility or pre-event compensation. The matched-VIX placebo compares FOMC windows with non-FOMC windows on days with similar lagged VIX, so a Fed-specific premium requires an excess over matched days.

State measurement. VIX comparability has changed over time as short-dated option volume has grown. This matters mainly for the exploratory VIX-change test, which is outside the frozen specification.

Specification search. The specification, threshold, outcome window and inference procedure are fixed in the pre-analysis plan before any estimate on post-2019 data, and the holdout is run once. Tests outside the plan are labelled exploratory and cannot change the primary conclusion.

5 Results

6 Discussion

7 Conclusion

References

A Notation and Additional Material

A.1 Notation

All returns are in basis points unless stated. Meeting m indexes scheduled FOMC meetings; $t(m)$ is the meeting date and $t(m)^-$ the previous valid session.

A.2 Specifications

Primary:

$$r_m = a + b\,A_m + \varepsilon_m, \qquad H_1 : a + b > 0. \tag{1}$$

Secondary, reported alongside and not substituted:

$$r_m = a + c\,\mathrm{VIX}_m + \varepsilon_m. \tag{2}$$

Inference uses CR1 cluster-robust standard errors by $g(m)$, wild cluster bootstrap-t (999 draws, seed 1) for $b = 0$ and $a + b = 0$, and a matched-VIX permutation placebo against non-FOMC days (2000 draws, seed 2). With few clusters the bootstrap p-values are primary and asymptotic t statistics descriptive.

Symbol	Definition	Unit
P_{prev}	16:00 ET last price on the previous valid session	index points
P_{end}	Last price at the end of the pre-announcement window: 13:45 ET (14:00 ET release, from 2013-03-01 and throughout the holdout) or 14:00 ET (14:15 ET release, earlier)	index points
rf_m	Risk-free rate over the same interval (daily file)	bps
r_m	Outcome: $10^4 \left[\ln(P_{\text{end}}/P_{\text{prev}})\right] - rf_m$	bps
VIX_m	VIX close five trading days before the meeting date	VIX points
τ	Anxiety threshold, fixed at 15.07 (development-sample median)	VIX points
A_m	$\mathbf{1}\{\text{VIX}_m > \tau\}$, the anxious indicator	0/1
a	Intercept: premium in calm meetings	bps
b	Anxious minus calm difference	bps
$a + b$	Premium in anxious meetings (primary test statistic)	bps
c	Slope of r_m on VIX_m in the secondary specification	bps per point
ε_m	Disturbance	bps
$g(m)$	Cluster (calendar year, a provisional proxy for volatility episodes)	—
G	Number of clusters (about seven in the holdout)	—

Table 1: Variables and parameters.

A.3 Hypotheses

H1 $a + b > 0$ in the holdout.

H2 The holdout anxious premium is below the development estimate of 19.3 bps.

H3 $b > 0$.

A.4 Sample definitions

Development sample: scheduled meetings from 2010-06 to 2019-12-31. Holdout: scheduled meetings from 2020-01-01. Unscheduled meetings are excluded. A meeting is usable if the meeting day and previous session are valid (no half-day, missing-trade or cash-closed flag) and r_m and VIX_m are non-missing. Release times are a calendar rule, not observed per meeting (UNVERIFIED).